

VERIFICATION OF WSL INSTALLATION:

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sai keerthana>wsl --status
Default Distribution: Ubuntu
Default Version: 2
WSL1 is not supported with your current machine configuration.
Please enable the "Windows Subsystem for Linux" optional component to use WSL1.

C:\Users\sai keerthana>
```

VERIFICATION OF GCC VERSION:

```
root@keerthana: ~
root@keerthana:~# gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

GCC PROGRAM:

```
GNU nano 7.2 process.c *
printf("\nChild Process\n");
printf("Child PID : %d\n", getpid());
printf("Parent PID: %d\n", getppid());

execlp(command, command, NULL);

// Executes only if command is invalid
printf("\nInvalid command!\n");
exit(1);
}

printf("\nChild process completed.\n");

return 0;
}
```

GNU nano 7.2

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo	M-6 Copy

```

root@keerthana:~# nano process.c
root@keerthana:~# gcc process.c -o process
root@keerthana:~# ./process
Enter a Linux command: pwd

Parent Process
Parent PID : 785
Child PID : 788

Child Process
Child PID : 788
Parent PID: 785
/root

Child process completed.
root@keerthana:~# |

```

SIMPLE COMMANDS:

```

root@keerthana: ~/OSLab
root@keerthana:~# mkdir OSLab
root@keerthana:~# ls
OSLab SKILL_WEEK3 exec_program.c process.c
OSSP commands.txt process
root@keerthana:~# ls
OSLab SKILL_WEEK3 exec_program.c process.c
OSSP commands.txt process
root@keerthana:~# ls -l
total 36
drwxr-xr-x 2 root root 4096 Aug 8 08:58 OSLab
drwxr-xr-x 11 root root 4096 Aug 3 05:13 OSSP
drwxr-xr-x 2 root root 4096 Aug 7 08:24 SKILL_WEEK3
-rw-r--r-- 1 root root 1467 Aug 1 08:02 commands.txt
-rw-r--r-- 1 root root 0 Aug 1 09:15 exec_program.c
-rwxr-xr-x 1 root root 16360 Aug 3 06:07 process
-rw-r--r-- 1 root root 934 Aug 3 06:07 process.c
root@keerthana:~# ls -a
. .cache .profile SKILL_WEEK3 process.c
.. .config .ssh commands.txt
.bash_history .local OSLab exec_program.c
.bashrc .notd_shown OSSP process
root@keerthana:~# cd OSLab
root@keerthana:~/OSLab# pwd
/root/OSLab
root@keerthana:~/OSLab# cd ..
root@keerthana:~# pwd
/root
root@keerthana:~# mkdir Test
root@keerthana:~# ls
OSLab SKILL_WEEK3 commands.txt process
OSSP Test exec_program.c process.c
root@keerthana:~# mkdir Test
root@keerthana:~# ls

```

```

root@keerthana: ~/OSLab
OSSP commands.txt process
root@keerthana:~# touch hello.c
root@keerthana:~# find . -name "*.c"
./hello.c
./SKILL_WEEK3/escape.c
./SKILL_WEEK3/task3.c
./SKILL_WEEK3/task1.c
./SKILL_WEEK3/task5.c
./SKILL_WEEK3/task2.c
./exec_program.c
./OSSP/task1.c
./OSSP/tests/test_parser.c
./OSSP/tests/test_executor.c
./OSSP/src/executor.c
./OSSP/src/builtin.c
./OSSP/src/fork_program.c
./OSSP/src/shell.c
./OSSP/src/exec_program.c
./OSSP/src/main.c
./OSSP/src/parser.c
./process.c
root@keerthana:~# locate hello.c
Command 'locate' not found, but can be installed with:
apt install plocate
root@keerthana:~# sudo apt update
sudo apt install plocate
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

```

```

root@keerthana: ~/OSLab
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  liburing2
The following NEW packages will be installed:
  liburing2 plocate
0 upgraded, 2 newly installed, 0 to remove and 71 not upgraded.
Need to get 160 kB of archives.
After this operation, 623 kB of additional disk space will be used.
Do you want to continue? [Y/n] sudo updatedb
Abort.
root@keerthana:~# y
y: command not found
root@keerthana:~# mkdir -p ~/OSLab
cd ~/OSLab
root@keerthana:~/OSLab# touch file1.txt
root@keerthana:~/OSLab# ls
file1.txt
root@keerthana:~/OSLab# echo "Hello Ubuntu" > file1.txt
root@keerthana:~/OSLab# cat file1.txt
Hello Ubuntu
root@keerthana:~/OSLab# cp file1.txt file2.txt
root@keerthana:~/OSLab# ls
file1.txt  file2.txt
root@keerthana:~/OSLab# mv file2.txt newfile.txt
root@keerthana:~/OSLab# ls
file1.txt  newfile.txt
root@keerthana:~/OSLab# rm newfile.txt
root@keerthana:~/OSLab# ls
file1.txt
root@keerthana:~/OSLab# printf "Line 1\nLine 2\nLine 3\nLine 4\nLine 5\nLine 6\nLine 7\nLine 8\nLine 9\nLine 10\n" > data.txt

```

```

root@keerthana: ~/OSLab
root@keerthana:~/OSLab# head data.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
root@keerthana:~/OSLab# tail data.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
root@keerthana:~/OSLab# more data.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10

```

```

root@keerthana: ~/OSLab
root@keerthana:~/OSLab# less data.txt
root@keerthana:~/OSLab# wc data.txt
10 20 71 data.txt
root@keerthana:~/OSLab# cp data.txt data2.txt
root@keerthana:~/OSLab# echo "Extra line" >> data2.txt
root@keerthana:~/OSLab# diff data.txt data2.txt
10a11
> Extra line
root@keerthana:~/OSLab# printf "banana\napple\norange\nmango\n" > fruits.txt
root@keerthana:~/OSLab# sort fruits.txt
apple
banana
mango
orange
root@keerthana:~/OSLab# printf "apple\napple\nbanana\nbanana\norange\n" > duplicate.txt
root@keerthana:~/OSLab# uniq duplicate.txt
apple
banana
orange
root@keerthana:~/OSLab# grep apple fruits.txt
apple
root@keerthana:~/OSLab# nano hello.c
root@keerthana:~/OSLab# cat hello.c
#include <stdio.h>

int main()
{
    printf("Hello Ubuntu!\n");
    return 0;
}
root@keerthana:~/OSLab# vi hello.c
root@keerthana:~/OSLab# gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0

```

```

root@keerthana: ~/OSLab
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

root@keerthana:~/OSLab# gcc hello.c -o hello
root@keerthana:~/OSLab# ls
data.txt  duplicate.txt  fruits.txt  hello.c
data2.txt  file1.txt      hello
root@keerthana:~/OSLab# ./hello
Hello Ubuntu!
root@keerthana:~/OSLab# which gcc
/usr/bin/gcc
root@keerthana:~/OSLab# whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz
root@keerthana:~/OSLab# man gcc
root@keerthana:~/OSLab# ps
  PID TTY          TIME CMD
  302 pts/0      00:00:00 bash
 1292 pts/0      00:00:00 ps
root@keerthana:~/OSLab# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0  08:57 ?            00:00:00 /sbin/init
root           2        1  0  08:57 hvc0       00:00:00 /init
root           6        2  0  08:57 hvc0       00:00:00 plan9 --cont
root          52        1  0  08:57 ?          00:00:00 /usr/lib/sys
root         100        1  0  08:57 ?          00:00:00 /usr/lib/sys
systemd+     111        1  0  08:57 ?          00:00:00 /usr/lib/sys
systemd+     112        1  0  08:57 ?          00:00:00 /usr/lib/sys
root         172        1  0  08:57 ?          00:00:00 /usr/sbin/cr
message+     173        1  0  08:57 ?          00:00:00 @dbus-daemon
root         180        1  0  08:57 ?          00:00:00 /usr/lib/sys
root         183        1  0  08:57 ?          00:00:00 /usr/libexec
syslog       195        1  0  08:57 ?          00:00:00 /usr/sbin/rs

```

```

root@keerthana: ~/OSLab
root      301      300  0  08:57 ?            00:00:00 /init
root      302      301  0  08:57 pts/0          00:00:00 -bash
root      303        2  0  08:57 pts/1          00:00:00 /bin/login -
root      346        1  0  08:57 ?            00:00:00 /usr/lib/sys
root      347      346  0  08:57 ?            00:00:00 (sd-pam)
root      371      303  0  08:57 pts/1          00:00:00 -bash
polkitd    608        1  0  08:59 ?            00:00:00 /usr/lib/pol
root      1038       1  0  09:13 ?            00:00:00 /usr/libexec
root      1293      302  0  09:18 pts/0          00:00:00 ps -ef
root@keerthana:~/OSLab# top
top - 09:18:11 up 21 min, 1 user, load average: 0.00, 0.00, 0
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 z
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 st
MiB Mem : 7796.6 total, 6833.2 free, 525.0 used, 594.
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7271.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	21860	13036	9676	S	0.0	0.2
2	root	20	0	3180	2200	2072	S	0.0	0.0
6	root	20	0	3196	2132	2008	S	0.0	0.0
52	root	19	-1	50456	16316	15200	S	0.0	0.2
100	root	20	0	25416	6700	5124	S	0.0	0.1
111	systemd+	20	0	21344	13164	11040	S	0.0	0.2
112	systemd+	20	0	91036	7984	7012	S	0.0	0.1
172	root	20	0	4244	2680	2432	S	0.0	0.0
173	message+	20	0	9672	5328	4588	S	0.0	0.1
180	root	20	0	17984	8884	7844	S	0.0	0.1
183	root	20	0	1756104	13148	10976	S	0.0	0.2
195	syslog	20	0	222516	5996	4632	S	0.0	0.1
203	root	20	0	3124	1972	1836	S	0.0	0.0
209	root	20	0	107032	23152	13696	S	0.0	0.3
300	root	20	0	3184	1104	980	S	0.0	0.0
301	root	20	0	3200	1244	1108	S	0.0	0.0

```

root@keerthana: ~/OSLab

```

302	root	20	0	6080	5356	3656	S	0.0	0.1
303	root	20	0	6676	4508	3736	S	0.0	0.1
346	root	20	0	20316	11568	9448	S	0.0	0.1
347	root	20	0	21168	3496	1784	S	0.0	0.0
371	root	20	0	6080	5288	3656	S	0.0	0.1
608	polkitd	20	0	308172	8144	7244	S	0.0	0.1
1038	root	20	0	370092	21092	18104	S	0.0	0.3
1294	root	20	0	9284	5704	3540	R	0.0	0.1
1295	root	20	0	25420	3864	2256	S	0.0	0.0

```

root@keerthana:~/OSLab# ps
  PID TTY          TIME CMD
  302 pts/0      00:00:00 bash
 1297 pts/0      00:00:00 ps
root@keerthana:~/OSLab# kill 1234
-bash: kill: (1234) - No such process
root@keerthana:~/OSLab# kill -9 1234
-bash: kill: (1234) - No such process
root@keerthana:~/OSLab# sleep 100 &
[1] 1301
root@keerthana:~/OSLab# jobs
[1]+  Running                  sleep 100 &
root@keerthana:~/OSLab# sleep 100
^Z
[2]+  Stopped                  sleep 100
root@keerthana:~/OSLab# jobs
[1]-  Running                  sleep 100 &
[2]+  Stopped                  sleep 100
root@keerthana:~/OSLab# bg %1
-bash: bg: job 1 already in background
root@keerthana:~/OSLab# fg %1
sleep 100
^C
root@keerthana:~/OSLab# sleep 60 &

```

```

root@keerthana: ~/OSLab
root@keerthana:~/OSLab# jobs
[2]+  Stopped                  sleep 100
[3]-  Running                  sleep 60 &
root@keerthana:~/OSLab# nano script.sh
root@keerthana:~/OSLab# ./script.sh
-bash: ./script.sh: Permission denied
root@keerthana:~/OSLab# chmod +x script.sh
root@keerthana:~/OSLab# ./script.sh
Hello from script
root@keerthana:~/OSLab# chmod 755 script.sh
root@keerthana:~/OSLab# ls -l script.sh
-rwxr-xr-x 1 root root 37 Aug  8 09:19 script.sh
root@keerthana:~/OSLab# touch file1
[3]-  Done                    sleep 60
root@keerthana:~/OSLab# ln file1 file2
root@keerthana:~/OSLab# ls -l
total 44
-rw-r--r-- 1 root root    71 Aug  8 09:02 data.txt
-rw-r--r-- 1 root root    82 Aug  8 09:04 data2.txt
-rw-r--r-- 1 root root    33 Aug  8 09:05 duplicate.txt
-rw-r--r-- 2 root root     0 Aug  8 09:20 file1
-rw-r--r-- 1 root root    13 Aug  8 09:01 file1.txt
-rw-r--r-- 2 root root     0 Aug  8 09:20 file2
-rw-r--r-- 1 root root    26 Aug  8 09:05 fruits.txt
-rwxr-xr-x 1 root root 15960 Aug  8 09:17 hello
-rw-r--r-- 1 root root    80 Aug  8 09:15 hello.c
-rwxr-xr-x 1 root root    37 Aug  8 09:19 script.sh
root@keerthana:~/OSLab# ln -s file1 shortcut
root@keerthana:~/OSLab# ls -l
total 44
-rw-r--r-- 1 root root    71 Aug  8 09:02 data.txt
-rw-r--r-- 1 root root    82 Aug  8 09:04 data2.txt
-rw-r--r-- 1 root root    33 Aug  8 09:05 duplicate.txt

```

```

root@keerthana: ~/OSLab
-rw-r--r-- 1 root root    13 Aug  8 09:01 file1.txt
-rw-r--r-- 2 root root     0 Aug  8 09:20 file2
-rw-r--r-- 1 root root    26 Aug  8 09:05 fruits.txt
-rwxr-xr-x 1 root root 15960 Aug  8 09:17 hello
-rw-r--r-- 1 root root    80 Aug  8 09:15 hello.c
-rwxr-xr-x 1 root root    37 Aug  8 09:19 script.sh
lrwxrwxrwx 1 root root     5 Aug  8 09:20 shortcut -> file1
root@keerthana:~/OSLab# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0 3.9G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K 3.9G  1% /mnt/wsl
drivers          476G  128G 348G 27% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.8G 954G  1% /
none            3.9G  40K 3.9G  1% /mnt/wslg
none            3.9G  0 3.9G  0% /usr/lib/wsl/lib
rootfs          3.9G  2.8M 3.8G  1% /init
none            3.9G  520K 3.9G  1% /run
none            3.9G  0 3.9G  0% /run/lock
none            3.9G  0 3.9G  0% /run/shm
none            3.9G  80K 3.9G  1% /mnt/wslg/versions.txt
none            3.9G  80K 3.9G  1% /mnt/wslg/doc
C:\             476G  128G 348G 27% /mnt/c
tmpfs           780M   20K 780M  1% /run/user/0
root@keerthana:~/OSLab# du -sh .
48K
root@keerthana:~/OSLab# free -h
              total        used         free       shared  buff/cache   available
Mem:          7.6Gi          525Mi        6.7Gi          3.5Mi          594Mi        7.1Gi
Swap:         2.0Gi           0B          2.0Gi
root@keerthana:~/OSLab# id
uid=0(root) gid=0(root) groups=0(root)
root@keerthana:~/OSLab# env
SHELL=/bin/bash

```

```

root@keerthana: ~/OSLab
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=keerthana
PWD=/root/OSLab
LOGNAME=root
HOME=/root
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/301_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*~=00;90:*#=00;90:*.*bak=00;90:*.*crdownload=00;90:*.*dpg=00;90:*.*dpg-dist=00;90:*.*dpg-new=00;90:*.*dpg-old=00;90:*.*dpg-tmp=00;90:*.*old=00;90:*.*orig=00;90:*.*part=00;90:*.*rej=00;90:*.*rpmnew=00;90:*.*rpmorig=00;90:*.*rpm-save=00;90:*.*swp=00;90:*.*tmp=00;90:*.*ucf-dist=00;90:*.*ucf-new=00;90:*.*ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=root
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/0
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/flatpak/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/C

```

```

root@keerthana: ~/OSLab
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=root
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/0
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/flatpak/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/sai keerthana/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/sai keerthana/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/sai keerthana/Downloads/38833FF26BA1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15_1_0_gdb_16_3_90_20250511_binutils_2_44_mingw_w64_v12_0_0/bin:/mnt/c/Program Files/Java/jdk-25:/mnt/c/Users/sai keerthana/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/sai ke
erthana/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/0/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/env
OLDPWD=/root
root@keerthana:~/OSLab# export PATH=$PATH:
root@keerthana:~/OSLab# export PATH=$PATH:
root@keerthana:~/OSLab#

```

```

root@keerthana: ~/OSLab
i keerthana/Downloads/38833FF26BA1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15_1_0_gdb_16_3_90_20250511_bin:/mnt/c/Program Files/Java/jdk-25:/mnt/c/Users/sai keerthana/AppData/Local/Programs/Microsoft VS
na/AppData/Roaming/npm:/snap/bin:..
root@keerthana:~/OSLab# alias ll='ls -l'
root@keerthana:~/OSLab# ll
total 44
-rw-r--r-- 1 root root 71 Aug 8 09:02 data.txt
-rw-r--r-- 1 root root 82 Aug 8 09:04 data2.txt
-rw-r--r-- 1 root root 33 Aug 8 09:05 duplicate.txt
-rw-r--r-- 2 root root 0 Aug 8 09:20 file1
-rw-r--r-- 1 root root 13 Aug 8 09:01 file1.txt
-rw-r--r-- 2 root root 0 Aug 8 09:20 file2
-rw-r--r-- 1 root root 26 Aug 8 09:05 fruits.txt
-rwxr-xr-x 1 root root 15960 Aug 8 09:17 hello
-rw-r--r-- 1 root root 80 Aug 8 09:15 hello.c
-rwxr-xr-x 1 root root 37 Aug 8 09:19 script.sh
lrwxrwxrwx 1 root root 5 Aug 8 09:20 shortcut -> file1
root@keerthana:~/OSLab# ls -l
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```